

# Day 5 — Why It Works — The Algebra

**Module:** Subtraction — Nikhilaṃ (All from 9) (Primary) · **Time:** 30 min

## Goal

Derive the place-value identity that makes the rule work.

## Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

## The flow

### Settle + chant (3 min)

Daily chant or breathing exercise.

### Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

### Today's idea (10 min)

- *Nikhilaṃ Navataścaramaṃ Daśataḥ* (IAST: Nikhilaṃ Navataścaramaṃ Daśataḥ; lit. “all from 9, the last from 10”)
- Why does the trick work? Let's see!

Write  $100 = 99 + 1$ . -  $99 - 37$  is easy (no borrow): 6 from 9 = 6 and  $9 - 3 = 6$ , write it out as:  $9-3 = 6$ ,  $9-7 = 2$ ... actually let's keep it simple. -  $100 - 37 = 63$ . - And  $9-3 = 6$ ,  $10-7 = 3 \rightarrow 63$ .

The 9s help because there's no borrow. “*That's the magic!*”

### Activity (8 min)

A simplified version of *Complement Flashcards* (see activities/). Draw, sing, or act it out.

### Wrap-up (3 min)

Each child shares one word that describes today.

## Homework

(See homework.md.)

## Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.

- If energy is low, swap the activity for free drawing.